Palaeomagnetic Investigations of the Tertiary and Quaternary igneous rocks: IV The Tertiary volcanic rocks of the Tokaji mountains, Hungary

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With 6 figures and 3 tables

Zusammenfassung

Es wird über paläomagnetische Messungen an ungarischen Gesteinen berichtet, die von 23 Punkten in den Tokajer Bergen stammen.

Die Ergebnisse werden als magnetische Feldumkehrungen gedeutet, da sich weder für Selbstumkehrungen noch für Beziehungen zwischen den jeweiligen Oxidationszuständen der Ilmenite und Titanomagnetite und den magnetischen Daten Anhaltspunkte ergeben haben.

Vier magnetische Zonen entsprechen dieser stratigraphischen Abfolge; ein möglicher Vergleich dieser Zonen mit der magnetischen Zeitskala von HEIRTZLER u. a. als Kontrolle legt die Vermutung nahe, die Miozän-Pliozän-Grenze mit 11,75 Millionen Jahren anzugeben.

Einerseits ist es wahrscheinlich, daß das Untere Sarmat durch die Tokajer Vulkanite nicht vollständig erfaßt wird, andererseits kann eine Säkular-Variation als Deutungsmöglichkeit wahrscheinlich ausgeschlossen werden.

Die virtuelle Pol-Position mit 70.4 W, 82.8 N weicht nicht bedeutend vom theoretischen Dipol-Feld ab.

Abstract

Measurements are reported from samples collected at twenty three sites in the Tokaj mountains of north-eastern Hungary. The simplest interpretation of the results is in terms of geomagnetic field reversals, for there is no petrological or magnetic evidence of self reversal or of correlation of oxidation state and magnetic sign. Four magnetic zones satisfy the minimum stratigraphic requirements, speculative use of these zones against the HEIRTZLER et al. magnetic time scale as a control suggests locating the Miocene-Pliocene boundary at 11.75 my. However it seems probable that much of the lower Sarmatian is not represented in the Tokaj samples, although the scatter of individual readings suggest secular variation is averaged out. The virtual pole position at 70.4 W, 82.8 N is not significantly different from the theoretical dipole field.

Résumé

Des mesures paléomagnetiques ont été effectuées sur des roches hongroises qui proviennent de 23 endroits des montagnes Tokaï.

Les résultats sont interprétés comme des inversions du champ magnétique car

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il n'y a pas de raisons d'admettre ni d'auto-inversion ni de rapports entre les états d'oxidation des ilménites et des titanomagnétites et les données magnétiques.

Quatre zones magnétiques correspondent à cette suite stratigraphique; une comparaison possible de ces zones avec l'échelle du temps de HEIRTZLER et al. comme contrôle laisse supposer que la limite Miocène-Pliocène se situe à 11.75 millions années.

Il est probable d'une part qu'une grande partie du Sarmatien inférieur n'est pas représentée par ces roches volcaniques de Tokaï, d'autre part qu'une variation séculaire comme un interprétation possible peut être exclue avec grande probabilité.

La position virtuelle du pôle avec 70.4 ouest, 82.8 nord n'est pas très différente du champ théorique du dipole.

Краткое содержание

Сообщается о результатах палеомагнетических измерений пород, взятых из 23 точек горы Токаи, Венгрия. Данные указывают на изменение магнитного поля, т. к. здесь нет доказательств самоизменения, или связи между состояниями окисления ильменита и титаномагнетита, и данными измерения магнитных свойств. Приводятся четыре магнитные зоны, соответствующие их стратиграфическому расположению. Сравнение этих зон со шкалой Heirtzler'a, как с контролем, разрещает предположить, что граница миоцен-плиоцен должна лежать 11,75 миллионов лет тому назад. — С одной стороны вероятно, что эти вулканиты Токая не представляют полностью породы нижнего сармата, с другой стороны, вариацию вековых изменений можно исключить. Принятое положение полюса в 70,4 W, 82,8 N не на много отличается от теоретического положения диполя.

Introduction

The Tokaji mountains lies in the northeastern part of Hungary close to the Slovakian and Russian frontiers. Topographically it takes the form of a low lying range of hills forming the natural southern extension of the Slaske Pohorie which reach as far north as Prešov in eastern Slovakia. The hills have a smooth rounded profile, and whilst wooded, do not appear to be as densely forested as their Slovakian counterparts. To the east lies the Tisza river at whose confluence with the Bodrog, stands toe town of Tokaj at which point stands the Kopasz hill the southernmost extension of the Tokaji mountains.

The hills are traversed by an adequate system of secondary roads linking with the major roads which skirt the foothills; these permit relatively easy access to most parts of the region. Although the region is not industrialized to any extent, there are numerous quarries in which fresh material could be collected with ease. In many parts however sampling had to be carried out at natural outcrops.

After a preliminary survey of the area with Dr. P. SZEMERÉDY, the material on which this account is based was collected with the help of P. GYARMATI in the following spring of 1965. The progress of the investigation has been helped by the friendly interest and assistance of Professor Dr. L. EGYED and Drs. P. SZEMERÉDY and P. MÁRTON of the Geophysical Institute of the Eotvös Loránd University, Budapest. The petrographic in-

vestigation was facilitated by Prof. Dr. H. VON PLATEN (Mainz), and Prof. Dr. H. HELKE (Mainz). The chemical analyses were prepared by Frau STEINBRUNN (Mainz) whose help is greatfully acknowledged.

General geology

The vulcanicity of the Tokaji mountains cannot be divorced from overall regional vulcanicity whose products mantle the entire north-eastern part of the country and extend northwards into Slovakia. The volume of dominantly acidic tuffaceous ejectmenta is enormous and the Tokaji Hegyseg and its continuation in Slovakia merely represent the more resistant masses poking through the blanket of pyroclastic material. They are more resistant due to the occurrence of andesitic and acidic lavas and ignimbrites. The trend of the Tokaj mountains and their Slovakian equivalents the Slaske Pohorie according to MAHEL' (1968) follows the strike of the Hornad fault zone.

The vulcanicity in northern Hungary appears to show a time trend with the youngest volcanics, those of the Tokaji mountains ranging up into the Sarmatian while the volcanics further to the west range down into the Helvetian. In the Tokaj range precise dating of the volcanics is difficult, not only because of stratigraphic problems but also because much of the volcanic material consolidated at medium or shallow depths (500—1500 m) with surface flows the exception (PANTO, 1968). In the Tokaji Hegyseg with its rhyolitic-rhyodacitic association, the proportion of pyroclastic rocks to the total igneous product is of the order of 80%, which is much higher than in regions of purely andesitic volcanism.

Table 1. Summary of Stratigraphy.

Sarmatian	Uppermost Sarmatian — Pliocene transition, limnic freshwater sediments with "archaic" forms of Pliocene molluscs and ostracods.
	Upper Sarmatian development of lagoons varying in salinity from brackish to fresh water. Fauna of brackish water mollusca and ostracods. Probably a short time interval.
	Lower Sarmatian probably represents more than half of Sarmatian characterized by brackish water mollusca and foramenifera. Evidence of fluctuations in salinity but less pronounced than in upper Sarmatian.
	"Abra beds" a formation transitional between Tortonian and Sarmatian some evidence of emergence. Regarded in Slovakia as freshwater Tortonian, but in Hungary the mollusca are considered to be definitely Sarmatian.
Tortonian	Upper Tortonian sediments and pyroclasts with a rich fauna. Contains evidence of submarine volcanics.

The stratigraphic problems in the Sarmatian are acute. The Tortonian sea shallowed producing in the Sarmatian a series of lagoons with periodically varying salinities presumably related to small sea level changes. Consequently a variety of fauna develop ranging from marine through

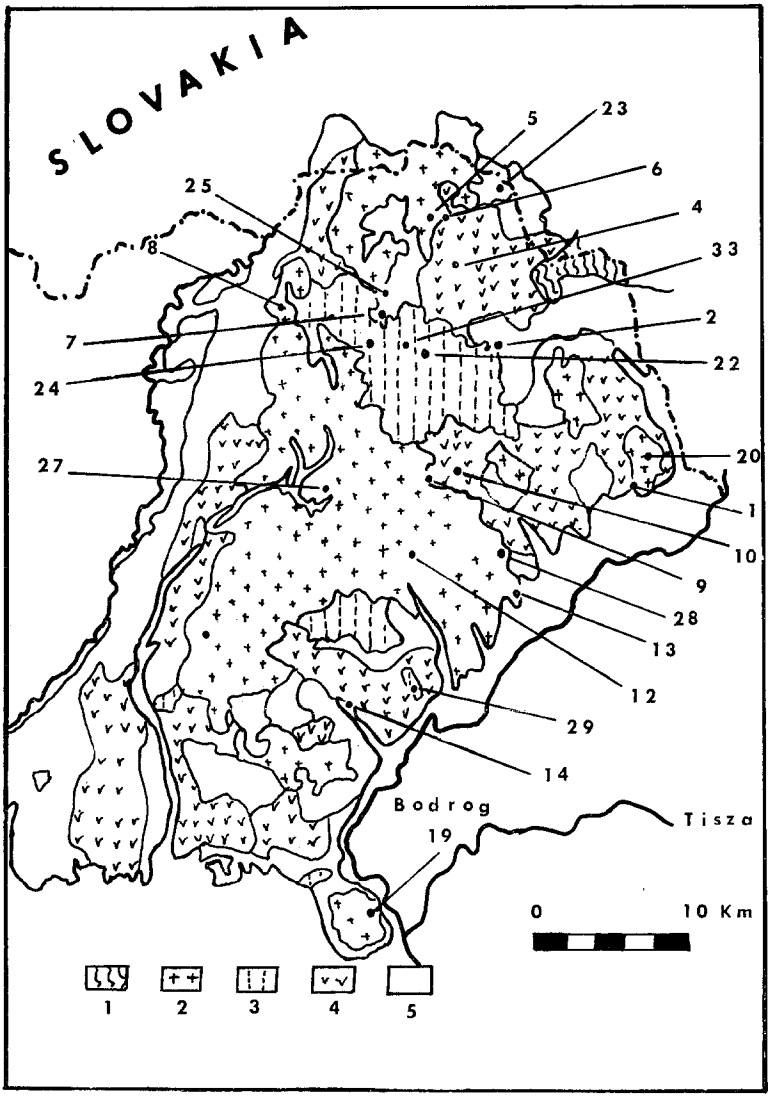


Fig. 1. Simplified geological sketch map of the Tokaj mountains. The site locations are numbered as in table 3. Map symbols: 1 = basement, 2 = andesite and dacite, 3 = rhyolitic ignispumite and perlite, 4 = welded rhyolitic ash flow, tuff, 5 = alluvial deposits, loess, etc.

brackish water to freshwater. These repeat as the appropriate facies repeat and it is in consequence difficult to define regional stratigraphic horizons on a time scale appropriate for palaeomagnetic purposes. What can be seen of the volcanics for any time increment is a considerable petrological

variety which means that the idealized petrographic sequence defined by Kuthan (1948, 1964) and Slavik in eastern Slovakia cannot be applied here, if indeed it is valid in Slovakia. The recently published radiometric dates from rocks in Central Slovakia (Konečný et al., 1969) suggest that the petrographic sequence does not apply there.

In the Tokaji mountains the oldest volcanics observed, submarine in type, are of Upper Tortonian age, the youngest are uppermost Sarmatian. The stratigraphic information for this time interval, according to the most recent interpretation, is summarized below (see Table 1, p. 729).

Petrological investigation

The majority of the samples collected were andesites, most of which were somewhat altered, others were rhyodacites and rhyolites as described by Panto and for 5 rocks this classification is born out by chemical analyses (table 2)[1]). It may be noted that the altered pyroxene andesite was weakly magnetized but could be shown to possess a reversed remanent magnetization.

In general the andesites possess a fine grained to glassy groundmass and all show some alteration of the coarser grained minerals. The hornblendes may be completely altered while nearly all the pyroxenes have an opacitic alteration rim in which ore minerals are developed. The pyroxenes may be chloritized, more rarely fresh, and commonly show intergrowths with the ore minerals.

Chloritization particularly of the groundmass is more marked in the rhyodacites, but here again ore minerals are found in alteration rims of the pyroxenes and also along cracks. The rhyolites show fluidal texture, and may be coloured by the development of haematite.

Of prime importance for palaeomagnetic work is the nature and oxidation state of the ore minerals. These are found not only as alteration products and intergrowths with pyroxenes but often very finely distributed in the groundmass. Good idiomorphic crystals are relatively rare. In the andesites the ore minerals may total 5—8% but as little as 1—3% by volume in the more acid rocks. The principal ore minerals present are ilmenite and titanomagnetite and their various oxidation products which can be classed according to the methods of Buddington and Lindsley (1964), Wilson and Haggerty (1966), Ade-Hall, Khan, Dagley and Wilson (1968) with three stages for the alteration of ilmenite and six for that of titano-magnetite. The low temperature oxidation with the production of maghemite could be observed in titanomagnetite classes 1—4. Maghemite first forms marginally on titanomagnetite and spreads in a net like form. Small amounts of pyrite is also observed.

Rather than describe each sample from each site individually, notes are attached to table 3, and the various oxidation stages illustrated by fig. 2 a—f. Examination of table 3 does not suggest any obvious correlation

[1]) In the description inclusive Table 3 the prefix-pheno-has to be added with the exception of the 5 rocks on Table 2.

Table 2. Chemical analyses with CIPW Norms
(locality names can be found by checking site numbers in table 3).

	Site 24 Wt. %	Site 23 Wt. %	Site 1 Wt. %	Site 2 Wt. %	Site 14 Wt. %
SiO_2	74,49	69,06	63,31	62,34	63,24
Al_2O_3	13,60	15,95	15,70	15,65	16,15
Fe_2O_3	0,90	2,46	4,10	2,90	2,33
FeO	0,41	0,29	0,95	2,70	2,99
MgO	0,16	0,81	2,44	3,35	0,81
CaO	1,57	1,68	5,18	5,95	3,58
Na_2O	3,30	3,30	2,70	2,50	4,00
K_2O	3,30	3,10	2,20	2,20	3,00
TiO_2	0,08	0,25	0,48	0,43	0,50
P_2O_5	0,03	0,03	0,09	0,06	0,06
H_2O^+	2,00	1,80	1,74	1,66	1,34
H_2O^-	0,10	1,30	1,03	0,22	0,18
MnO	0,03	0,01	0,13	0,13	0,12
	99,97	100,04	100,05	100,09	CO_2 1,76
					100,06
CIPW-Norm					
Qz	39,11	33,23	25,41	21,71	23,19
Or	19,50	18,32	14,10	13,00	17,73
Ab	27,95	27,95	22,87	21,18	33,88
An	7,59	8,18	21,21	24,98	6,30
Di	—	—	3,03	3,38	—
Hy	0,45	2,04	4,91	8,81	4,98
Il	0,15	0,48	0,91	0,82	0,95
Mt	1,09	0,21	1,82	4,21	3,38
Hm	0,15	2,32	2,80	—	—
Ap	0,07	0,07	0,21	0,14	0,14
C	1,80	4,15	—	—	4,0
Aq	2,10	3,10	2,77	1,87	1,51
CC	—	—	—	—	3,99
	99,96	100,05	100,04	100,10	100,05
	Rhyolitic Ignimbrite	Rhyo-dacite	pyroxene-dacite	pyroxene-dacite	altered pyroxene dacite

of magnetic sign with degree of oxidation. Normally magnetized rocks are found in both low and high oxidation states and the same is true in the case of reversely magnetized samples.

Palaeomagnetic investigation

The basic techniques used here are the same as those described in the reports of earlier studies (NAIRN, 1967; NAIRN & VOLLSTÄDT, 1968) and will in consequence only be briefly noted here.

a) Direction of magnetization. Although at a few sites, samples were obtained by coring with a portable drill, most samples obtained in the form of oriented blocks from which cores were sub-

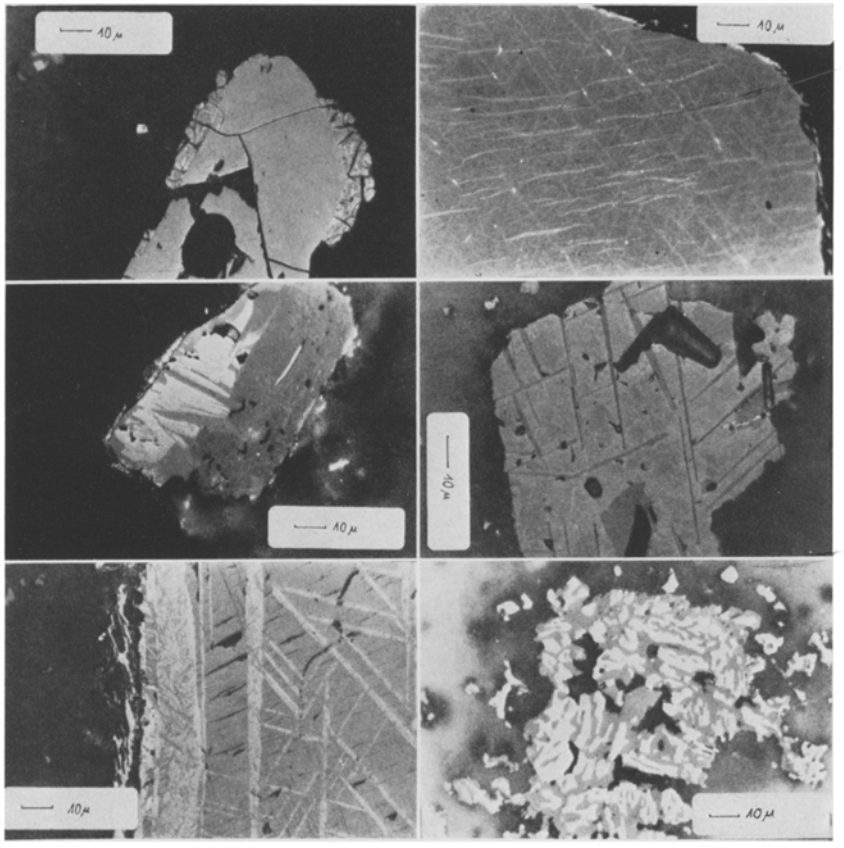


Fig. 2. Photomicrographs illustrating the various stages and oxidation forms found in the Tokaj volcanic rocks.

a) Ilmenite stage I with a rim of original titanomagnetite (specimen from site 2).
b) Ilmenite stage II with a rutile net (specimen from site 27).
c) Ilmenite stage III with pseudobrookite (dark grey) rutile (medium grey) and haematite (a light grey-white) (specimen from site 16).
d) Titanomagnetite class 2. Almost completely replaced by titanomaghemite (specimen from site 19).
e) Titanomagnetite class 5 containing a black spinal phase, and ilmenite which has been altered to an aggregate of haematite and rutile (medium grey) (specimen from site 12).
f) Irregular intergrowth of pseudobrookite and haematite (specimen from site 29).

sequently obtained in the laboratory. At each of twenty four sites approximately six samples were collected. No definite sampling system was used, samples were distributed to give an adequate representation of the outcrop. This might mean several metres between samples from a large outcrop, much less at less extensive outcrops.

For the most part the directions of remanence were measured under a simple astatic magnetometer on one core from each sample. Samples were progressively demagnetized after measurement of the natural remanence, in fields ranging up to 600 oe. or 1200 oe. In general the least dispersion

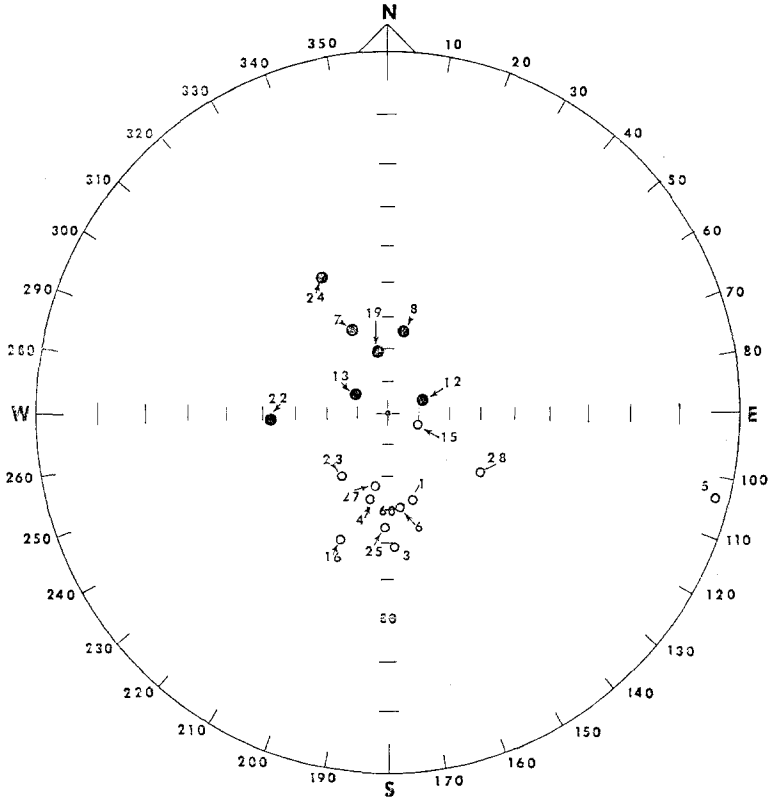


Fig. 3. Site mean directions of magnetization of rocks from the Tokaji Hegyseg. Following the usual conventions the north seeking pole pointing downwards is indicated by a solid point, and a reversed magnetization with the north seeking pole pointing upwards is represented by an open side. The sites are numbered and correspond to site numbers in the table summarizing the statistical data.

of the site mean direction occurred after demagnetization in fields of 77 or 150 oe. Site mean directions of magnetization were computed on a UNIVAC 1108 using the program devised by S. Grommé. When the least dispersion coincided with the NRM measurement the constraint was applied that during demagnetization there should be no significant direction change. A number of sites had intensities of magnetization too low for satisfactory results. The data is summarized in table 3 and fig. 3 which shows the site mean directions of magnetization.

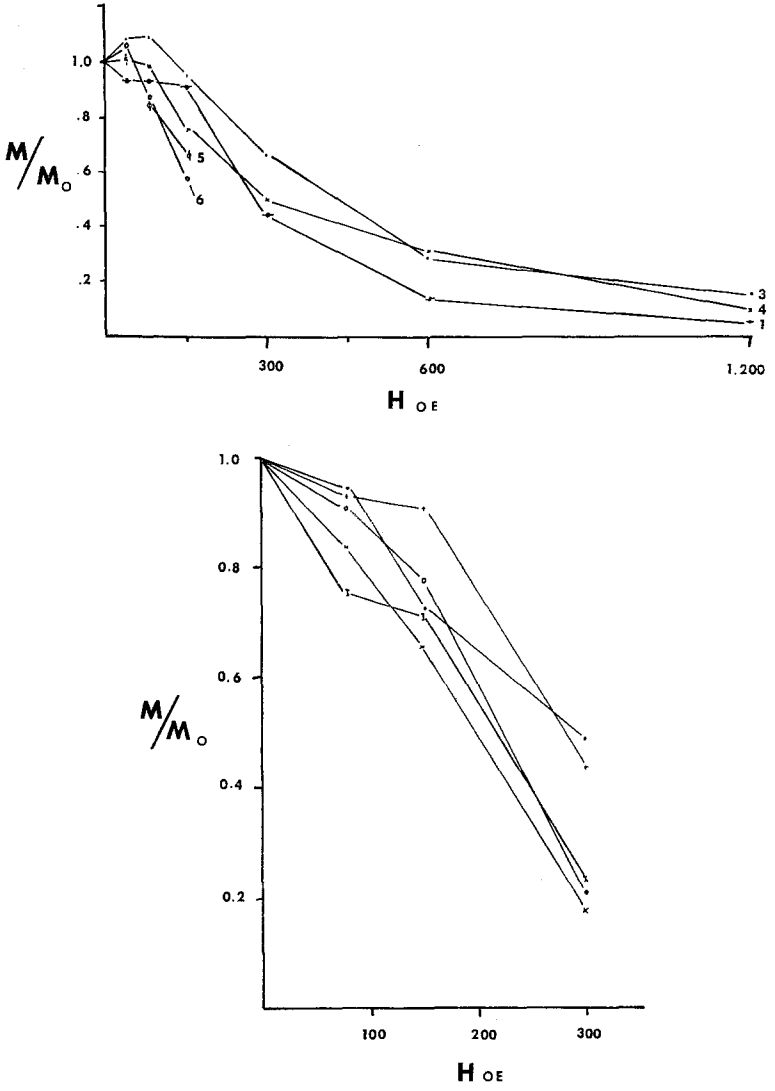


Fig. 4. Normalized demagmatization curves illustrating the change in intensity with increasing A. C. fields. The numbers correspond to the site from which the sample was drawn.

b) Intensity of magnetization. The measurement of intensity of magnetization was carried out concurrently with the direction of remanence. All samples were progressively demagnetized in A. C. fields while rotating about three axes. Normalized demagnetization curves for both normal and reversely magnetized rocks are shown in fig. 4. Changes in the direction of remanence during demagnetization can be studied

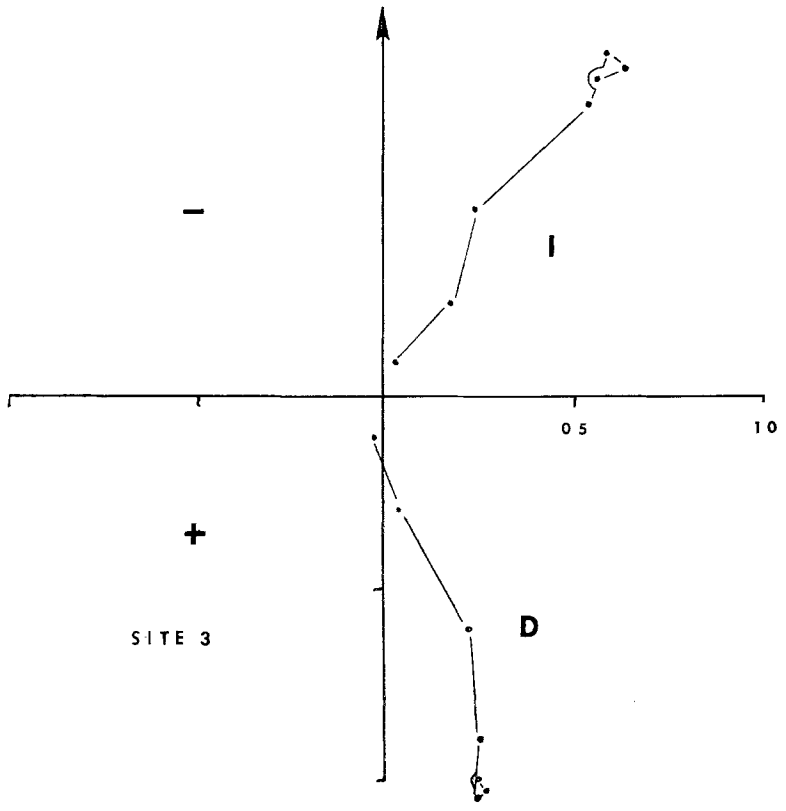


Fig. 5. The change of direction and inclination of magnetization with change in intensity of magnetization. Radial distance from the center of the diagram represents normalized intensity, declination is plotted in the horizontal plane, inclination in the vertical.

using the normalized intensity as a parameter. The curve shown in fig. 5 is difficult to interpret as the declination and inclination lines do not pass through the origin. This has been observed by other workers such as WILSON and ZIJDERVELD. In table 2 the site mean intensity of N.R.M. and of magnetization at minimum dispersion are both given logarithmic form, as intensity distribution appears to follow a log normal law (IRVING, MOLYNEUX, RUNCORN, 1966). This is illustrated in fig. 6 in which the cumulative frequency curve is based upon individual samples and not site means because of the wide distribution of intensities possible at any site.

Discussion of results

The directions of natural remanent magnetization are widely scattered and this is more apparent in the normally magnetized rocks than in those with reverse magnetization and this is still true after demagnetization.

There is one result which has anomalously low inclination (site 5) although the scatter of the measurements of individual samples is like the others. It must be regarded as real and it may represent an extreme excursion of the field during secular variation since the intensity of magnetization is comparable with other sites. The magnetic sign is given even for sites

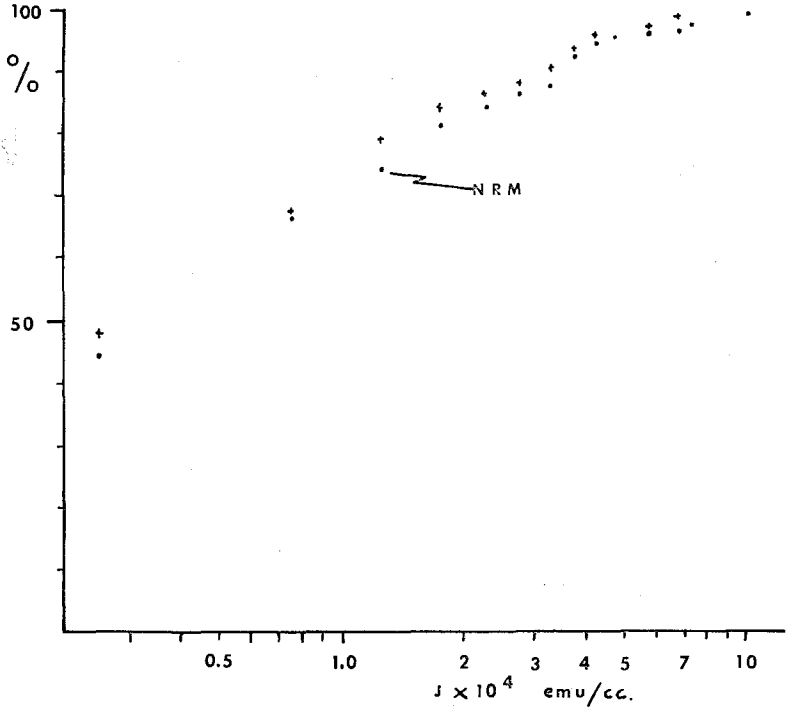


Fig. 6. Cumulative percentage curves showing the distribution of N.R.M. and of demagnetized intensities of rocks from the Tokaji hegyseg. The curves are drawn on semi-log paper.

whose intensity of magnetization was too low to permit satisfactory demagnetization experiments to be carried out.

The petrological data does not indicate any obvious correlation of magnetic sign with oxidation state, and in consequence it is considered that magnetic sign changes reflect reversals of the geomagnetic field and are thus of stratigraphic significance. The major handicap in attempting interpretation is the absence of any radiometric age control, and furthermore in rocks of this age it is not possible to distinguish events by K/A methods.

In an earlier paper (NAIRN, 1967) an attempt was made to establish a sequence of magnetic zones in Eastern Slovakia making use of the petrological model for the sequence of events established by KUTHAN (1968).

Table 3.

Site	Locality, Age, Lithology	Magnetization Declination	Magnetization Inclination	Sampling a/b	Confidence α 95	Dispersion K	Vector R	Demag. Field oe.	Summary of the ore mineralogy
19	Tokaj-Patko uppermost Sarmatian px dacite	349.9	+ 68.9	5/7	4.3	316	4.987	77	Titanomagnetite classes 1—6, max. class 5, Ilmenite stages I—III, max. stage II maghemite in titanomagnetite
8	Gönc-Csenko uppermost Sarmatian px amphibole dacite	9.6	+ 69.3	6/7	10.9	39	5.871	150	Titanomagnetite class 1, Ilmenite stage I Ilmenite much commoner than titanomagnetite
29	Olaszliszka uppermost Sarmatian „fluidal" rhyolite	(208.8	+ 80.0)*)	3/6	8.4	215	2.991	777	Irregular haematite-pseudobrookite intergrowths (Ilmenite stage III?) granulated titanomagnetite
23*	Puxztafalu-Tolvajhegy upper Sarmatian rhyodacite	218.5	— 64.6	6/6	5.0	183	5.973	77	Ilmenite stage II, biotite and hornblende mostly altered to opaque minerals
27	Regecs-Varhegy upper Sarmatian rhyodacite	188.8	— 65.4	5/6	11.5	46	4.912	150	Titanomagnetite classes 1, 4—5 (max.) in crystals, (1—5 in groundmass) in part granulated. Ilmenite stages I—III Ilmenite-titanomagnetite intergrowths
33	Rostallo L-U Sarmatian px amphibole dacite	177.8	— 47.9	4/6	8.4	120	3.975	77	Finely divided ore minerals in groundmass. Titanomagnetite class 4—5, often with haematite rims. Ilmenite stage II intergrown with titanomagnetite
4	Fuzerkomlos L-U Sarmatian px andesite	192.2	— 62.2	5/5	14.5	29	4.860	NRM	Titanomagnetite class 1 but mostly altered to a micaceous goethite. Ilmenite stages I—II. Some pyrite. Groundmass too finely divided for determination
6	Fuzer-Varhegy L-U Sarmatian px dacite	170.8	— 60.0	6/6	7.3	86	5.942	150	Titanomagnetite class 1—2, rarely 3, Ilmenite stage I in both larger crystals and groundmass. Ilmenite with titanomagnetite rims. Rare pyrite
14*	Erdobenye-Mulatohegy L-U Sarmatian carbonatized px andesite	too weak		two	samples	R	stable up	to 300	Titanomagnetite class 1—3, altered marginally and along cracks to a micaceous goethite. Ilmenite stages I—II intergrown with titanomagnetite

15	Aranyosvolgy z L-U Sarmatian px andesite	109.7	— 81.0	6/6	4.2	250	5.980	150	Little titanomagnetite mostly as a rim on ilmenite. Ilmenite stages I—III (max. stage III). Also finely divided in groundmass
5	Fuzer-Pivotka L-U Sarmatian px dacite	114.0	— 1.4	5/5	11.3	47	4.915	77	Titanomagnetite and Ilmenite intergrowths. Ilmenite (idiomorphic) stage I (rarely II) sometimes with haematite lamellae. Titanomagnetite class 1 but may be partially or completely altered to maghemite
22	Kemencepatak-Kokapu lower Sarmatian red rhyolite	268.0	+ 52.3	6/6	9.4	52	5.903	150	Few large grains, mostly as fine pigment. Ilmenite stage I (rarely stage II) sometimes includes haematite lamellae
24 *	Nagybozsva-Senyo valley lower Sarmatian rhyolitic ignimbrite	333.1	+ 42.6	5/6	7.0	92	5.946	NRM	Ore mostly finely divided in groundmass largely unidentifiable Ilmenite stages I—II
12	Komloska-Pusztavar lower Sarmatian px andesite	69.2	+ 78.3	6/6	9.6	50	5.900	150	Large ore grains as well as small groundmass particles. Titanomagnetite classes 1—6 (max. class 4 in which titanomaghemite is found) Ilmenite stages I—III (max. stage III)
9	Haromhuta lower Sarmatian welded rhyolitic ash flow	too weak		two	samples	N	stable up	to 150	Very little fine grained ore in groundmass Ilmenite stage II
7	Nagybozsva-Senyo volgy lower Sarmatian px andesite	334.8	+ 61.3	6/6	8.5	63	5.920	77	Little irregularly formed ore grains. Titanomagnetite class 5—6. Ilmenite stage 1 but II—III in groundmass. Titanomagnetite in groundmass class 1—3 haematite common, some goethite and lepidocrocite
25	Telkibanya-Matyas kiraly kutja Abra beds banded rhyodacite		+	3/5	—	—	—	77	Phenocrysts ore minerals rare. Titanomagnetite class 4—6, class 5 grains with residual titanomagnetite and spinel. Haematite in ilmenite region intergrowth with rutile. Ilmenite stage III. Haematite in and around mafic minerals

*) At 1200oe two samples are measureable, both normally magnetized.
*) Chemical analyses given in table 2. --- broken lines correspond to stratigraphic boundaries.

Table 3 cont.

Site	Locality, Age, Lithology	Magnetization Declination	Magnetization Inclination	Sampling a/b	Confidence α 95	Dispersion K	Vector R	Demag. Field oe.	Summary of the ore mineralogy
10	Harombuta-Ujhuta, Hangyaszo volgy Abra beds px amph. dacite	23.6	+ 45.4	4/6	8.8	109	3.973	77	Large irregular ore grains and fine groundmass particles. Ilmenite titanomagnetite intergrowths. Same idiomorphic grains. Titanomagnetite class 1 Ilmenite stage I. Haematite in mafic minerals and as rims on ore minerals
2*	Kishuta Abra beds px andesite	too weak		two	samples	N	stable to	150	Large grains usually associated with mafic minerals, small particles in groundmass. Ilmenite stage I. Titanomagnetite mostly in class 1, may be granulated or altered to maghemite.
13	Sarospatak-Szent Vince hegy Abra beds px andesite	300.9	+ 78.8	6/6	4.9	246	5.984	77	Ilmeniate intergrowths with titanomagnetite. Titanomagnetite class 1—3 but may be completely altered to maghemite. Ilmenite stages I—II (max. stage I), may contain titanomagnetite lamellae or have a titanomagnetite rim.
20	Satoraljaujhely-Nemahegy Abra beds px andesite	182.4	— 55.3	6/6	7.0	92	5.946	77	Ore in groundmass not identifiable. Titanomagnetite class 1. Ilmenite stages I—II. Titanomagnetite and haematite in mafic minerals
28	Hercegkutt-Gombos Abra beds px andesite	123.0	— 55.0	6/6	4.5	227	5.978	77	Ores either as large grains or as fine pigment in matrix. Titanomagnetite stage 1 in grains 1—5 (max. 5) in groundmass. Ilmenite stage I—II in both. Some pyrites
1*	Satoraljaujhely-Satorhegy upper Tortonian px dacite	165.5	— 61.0	6/6	6.4	109	5.954	300	Ore as large grains, unidentifiable in groundmass. Ilmenite stage I. Titanomagnetite class 1, and some haematite in mafic minerals.

That sequence is not upheld by the present work in which fewer magnetic zones were required, nor is it upheld in Central Slovakia by radiometric dating (KONEČNÝ et al., 1969).

When the East Slovakian results are examined in the light of the Hungarian measurements, it is found that whereas in Hungary, rhyolites are recorded in both lower and upper Sarmatian, in Slovakia they are all relegated to a single horizon in the lower Sarmatian. As both normal and reversely magnetized rhyolites were found, this required the interpolation of an extra zone. If rhyolites also occur at Slovakia at more than one level, as seems not unreasonable both in the light of the variety of rock types within a limited area, and in view of the wide distribution of ages found by radiometric work in central Slovakia, then the difference could be reconciled. Thus consideration of the Hungarian palaeomagnetic results alone, in a region where the stratigraphical control is better, leads to the conclusion that the volcanic history is more complex than the simple model of KUTHAN would lead one to suppose. The recent radiometric work of KONEČNÝ et al. (1969) on the volcanics and intrusives of the Central Slovakian province mentioned above clearly indicates that a simple petrological model does not apply there.

The simplest sequence of magnetic zones from the Tokaj mountains, consistent with the stratigraphy, comprises two normal and two reversed zones. Geological considerations imply that the two upper zones are relatively short. The lower of the two normal zones in the lower Sarmatian should then represent more than half Sarmatian times. Measurements were not made from rocks below the Abra beds. Although such a sequence of zones now matches the Japanese results from the Sendai province (NOMURA, 1953) and possibly the work of COOMBES & HATHERTON (1959) in New Zealand, it seems unlikely that this is more than coincidence.

The major problem facing the interpretation of the palaeomagnetic results in a more than local sense, is the uncertainty concerning the location of the Miocene-Pliocene boundary. This boundary has been variously placed at 7 my. (FUNNELL, 1964), 11 my. (HOLMES, 1959), 12 my. (EVERNDEN et al., 1961), and 13 my. $\pm$ 1 my. (KULP, 1961). If the equating of the Clarendonian with the Sarmatian (DURHAM, JAHNS, SAVAGE, 1954) is reasonable, then a figure of approximately 10 my. can be added. In the data summary of KONEČNÝ et al. (1969, fig. 1) the Miocene-Pliocene boundary is placed at 12 my. with the duration of the Sarmatian given as 3 my. If the simple sequence of magnetic zones for the Tokaj mountains is compared against the magnetic zonal sequence given by HEIRTZLER et al. (1968), then the position which seems to fit best, providing two short periods above a longer normal interval within the given range of ages of the Miocene-Pliocene boundary and consistent with the local stratigraphic data (table 1) gives a figure of around 11.75 my. for that boundary. However the duration of the Sarmatian given by the Tokaj magnetic zones, from 11.75 my. to around 12.5 my. seems too short, although the 3 my. of KONEČNÝ et al. (1969) is not necessarily correct.

If we may conclude that the two short zones at the top of the Sarmatian sequence (see table 1) suffice to define the absolute position of the Miocene Pliocene boundary by their magnetic signature, then it seems probable that much of the remainder of the Sarmatian is either poorly represented, or is not represented at all, as the alternative to the period being as short as the model of zones would imply (0.75 my. for the four zones compared to the ten zones which would be anticipated if the Sarmatian were truly of 3 my. duration). The history of vulcanicity in the region is of sporadic outbursts of volcanic activity of short duration, and so the most reasonable conclusion is that the volcanic horizons represent only a small portion of the lower Sarmatian, and in consequence comparisons with other regions serves no useful function at the present stage.

A mean pole position was computed based upon seventeen of the sites tabulated (excluding FUZER-PIVOTKA and OLASZLISZKA), the data is given below:

Virtual pole N 70.4 W 82.8 N dp = 11.9 dm = 14.6.

It was obtained by reversing the sign and declination of reversely magnetized samples giving the following overall mean direction

$$349.2 + 65.6 \quad R = 16.041 \quad K = 17 \quad \alpha_{95} = 9.0.$$

Conclusions

A sequence of four magnetic zones satisfies the local stratigraphic succession. This scheme while superficially in agreement with upper Miocene successions elsewhere is probably an incomplete representation of that time interval. The uppermost beds seem to provide a magnetic signature for the location of the Miocene-Pliocene boundary at 11.75 my. This stratigraphic interpretation of the magnetic data is regarded as valid for petrological examination indicates that there are no obvious grounds for associating magnetic sign with oxidation state.

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